

Security analysis and numerical simulation

1. Gate minimization using exclusive-or-sum-of-products (ESOP)

Exclusive-or sum-of-products (ESOP) is a two-level Boolean representation in which product terms are combined using XOR operations. ESOP minimization is particularly important in quantum computing because it can reduce the number of multi-controlled NOT (n-CNOT) gates. When multiple n-CNOT gates share the same target qubit, their combined effect can be represented as an ESOP expression, with each control set corresponding to a product term. In the NEQR framework [1], the Espresso algorithm [2] is used to minimize sum-of-products (SOP) expressions and reduce the number of n-CNOT gates. However, the resulting SOP expression must be converted into an ESOP form before implementation, and a minimal SOP does not necessarily yield a minimal ESOP. Therefore, direct ESOP minimization is often more effective, as it produces optimized ESOP expressions without requiring additional conversion. A comparison of ESOP minimization and the Espresso algorithm is provided in [3].

Consider the unitary operation of a general n -CNOT gate acting on n control qubits and a single target qubit. Let this operator be denoted by V_ϕ and its corresponding ESOP term be U_ϕ . The control configuration is represented by an n -bit binary sequence ϕ containing don't-care entries (*). Each position in ϕ specifies whether the corresponding qubit acts as a positive control, a negative control, or has no influence on the target qubit. If ϕ contains m don't-care positions, replacing each (*) with either 0 or 1 generates a set Φ of 2^m binary strings.

$$\begin{aligned}\phi &= \phi^0 \phi^1 \dots \phi^{n-1}, \quad \phi^i \in \{0, 1, *\} \quad 0 \leq i < n. \\ \Phi &= \{s \in \{0, 1\}^n \mid (\phi^i = *) \vee (\phi^i = s^i), 0 \leq i < n\}.\end{aligned}\tag{1}$$

The operator V_ϕ and the product term U_ϕ can be described as

$$V_\phi = \sum_{\substack{P=0 \\ P \notin \Phi}}^{2^n-1} I \otimes |P\rangle\langle P| + \sum_{\substack{P=0 \\ P \in \Phi}}^{2^n-1} X \otimes |P\rangle\langle P|,\tag{2}$$

$$U_\phi = \prod_{i=0}^{n-1} \nu_i, \quad \text{where } \nu_i = \begin{cases} v_i, & \phi_i = 1, \\ \bar{v}_i, & \phi_i = 0, \\ 1, & \phi_i = *. \end{cases} \quad (3)$$

The minimization procedure is as follows. Let us assume we have an operation composed a sequence of q multi-control COT gates, $F = \prod_{i=0}^{q-1} V_{\phi_i}$, operating on the same set of qubits. This corresponds to ESOP function $f = \bigoplus_{i=0}^{q-1} U_{\phi_i}$. Using ESOP minimization algorithms, the function can be minimized to get an equivalent form with t product terms, $f = \bigoplus_{i=0}^{t-1} U_{\phi_i}$. This expression can be used to construct the minimized quantum operation $F = \prod_{i=0}^{q-1} V_{\psi_i}$.

1.1. Illustrative quantum simulation of the proposed algorithm for a 4×4 matrix

A demonstration of the proposed algorithm on a quantum simulator utilizing a 4×4 quantum image is provided for verification and validation. The quantum circuit is simulated with the StatevectorSimulator provided by Qiskit Aer on a device featuring an Intel i9 11th Gen, 16 GB RAM. To enhance clarity, only one iteration of the encryption procedure is illustrated, use a single S-box for the complete matrix. Additionally, the $m - CNOT$ gates are optimized via the EXORCISM-4 [4, 5] algorithm at every level. A demonstration of the proposed algorithm on a quantum simulator utilizing a 4×4 quantum image is provided for verification and validation. Table 1 illustrates a comparison of circuit depths achieved with and without the implementation of EXORCISM-4.

Table 1: Circuits depth with and without ESOP optimization for the 4×4 matrix.

Unitary operators	Without ESOP optimization	With ESOP optimization
NEQR encoding	89	28
QAT	4	4
CSGC	15	15
Chaotic XOR diffusion	62	33
Substitution box	2014	645

1.1.1. NEQR preparation

The plain image C , given in Eq.(4), is encoded in NEQR to create the quantum plain image. The location information is stored in the registers y and x , and the pixel intensity is stored in the register F . Figure 1 shows the circuit to create the

quantum plain image. The qubits are in the state $|C\rangle$.

$$\begin{pmatrix} 0 & 15 & 61 & 166 \\ 63 & 188 & 18 & 111 \\ 135 & 99 & 207 & 133 \\ 191 & 21 & 223 & 200 \end{pmatrix} \quad (4)$$

The expression of the state $|C\rangle$ is

$$\begin{aligned} |C\rangle = \frac{1}{4} & (|00001100\rangle|0000\rangle + |00100010\rangle|0001\rangle + |00111000\rangle|0010\rangle + |01001110\rangle|0011\rangle \\ & + |01011010\rangle|0100\rangle + |01111011\rangle|0101\rangle + |10010001\rangle|0110\rangle + |10100111\rangle|0111\rangle \\ & + |11001001\rangle|1000\rangle + |11011110\rangle|1001\rangle + |00010001\rangle|1010\rangle + |00111111\rangle|1011\rangle \\ & + |01011000\rangle|1100\rangle + |00101100\rangle|1101\rangle + |10000100\rangle|1110\rangle + |11111010\rangle|1111\rangle) \end{aligned} \quad (5)$$

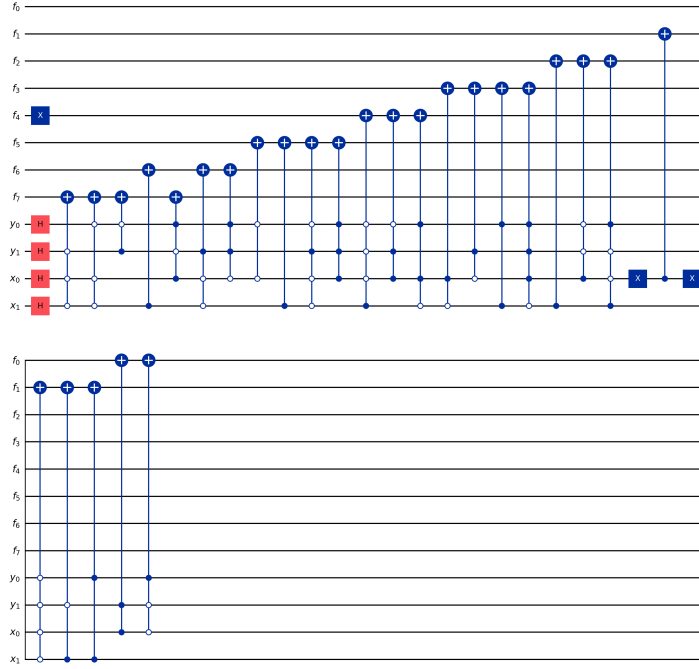


Figure 1: NEQR quantum circuit.

1.1.2. QAT confusion

The QAT operaotr is defined in the Eq.(6) is as follows:

$$\begin{cases} |u'\rangle = |(u + v) \bmod 2^m\rangle \\ |v'\rangle = |(u + 2v) \bmod 2^m\rangle \end{cases} \quad (6)$$

The state of the system becomes $|C'\rangle$, and can be described as :

$$\begin{aligned} |C'\rangle = \frac{1}{4} & (|00001100\rangle|0000\rangle + |00101100\rangle|0001\rangle + |00010001\rangle|0010\rangle + |10100111\rangle|0011\rangle \\ & + |00111111\rangle|0100\rangle + |01011010\rangle|0101\rangle + |00100010\rangle|0110\rangle + |10000100\rangle|0111\rangle \\ & + |00111000\rangle|1000\rangle + |11111010\rangle|1001\rangle + |11001001\rangle|1010\rangle + |01111011\rangle|1011\rangle \\ & + |11011110\rangle|1100\rangle + |10010001\rangle|1101\rangle + |01001110\rangle|1110\rangle + |01011000\rangle|1111\rangle) \end{aligned} \quad (7)$$

Fig. 2,3,4, 5, 6, 7, 8, 9 depicts the Arnold scrambling quantum circuit for $n = 6$.

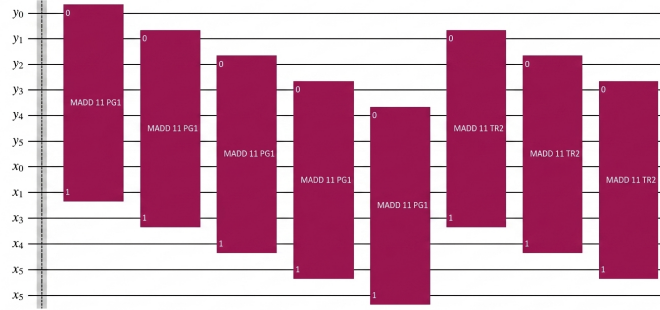


Figure 2: Arnold quantum circuit (a of n).

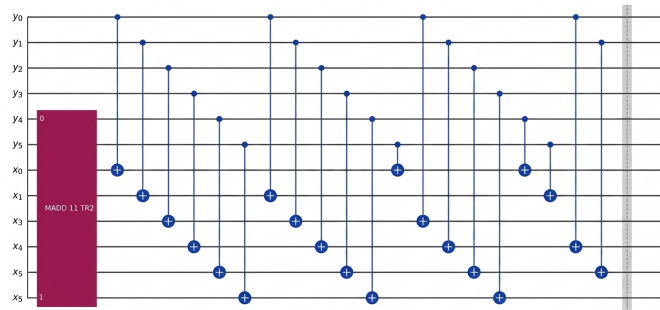


Figure 3: Arnold quantum circuit (b of n).

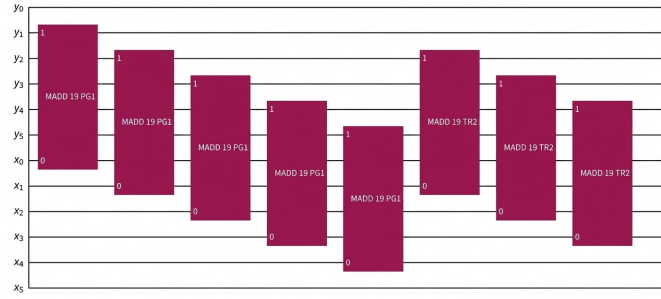


Figure 4: Arnold quantum circuit (c of n).

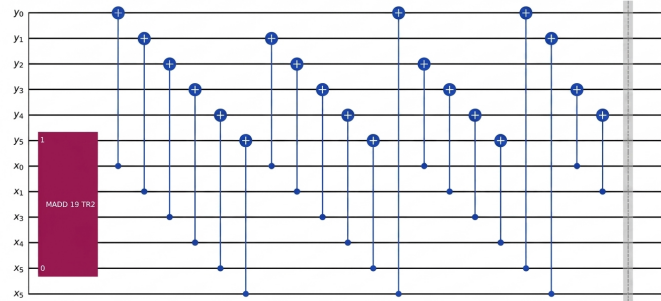


Figure 5: Arnold quantum circuit (d of n).

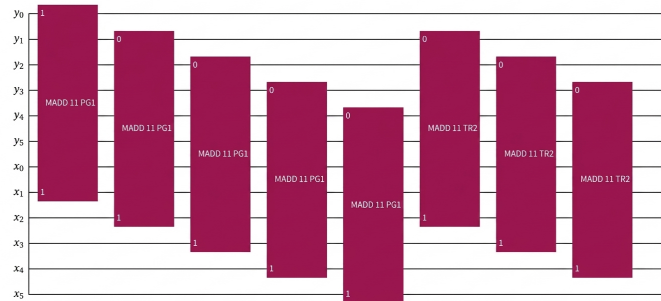


Figure 6: Arnold quantum circuit (e of n).

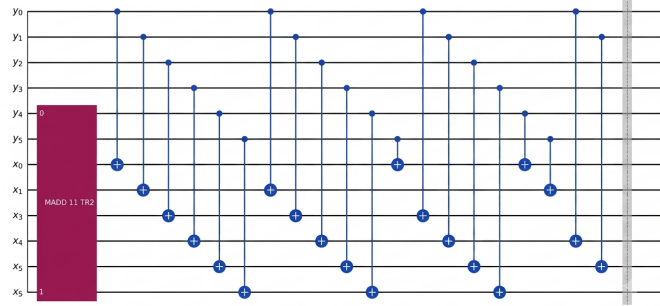


Figure 7: Arnold quantum circuit (f of n).

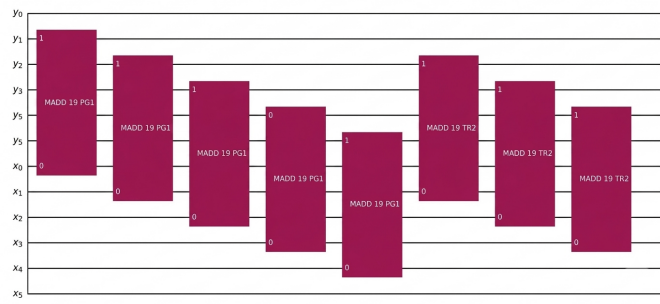


Figure 8: Arnold quantum circuit (g of n).

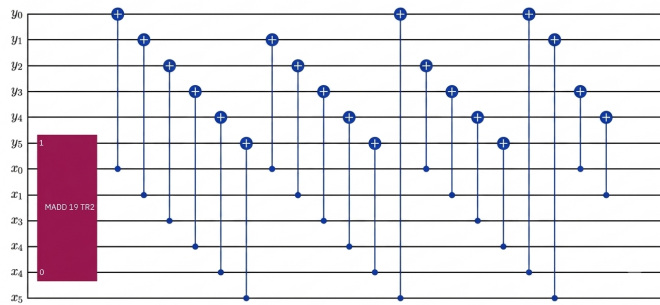


Figure 9: Arnold quantum circuit (h of n).

1.1.3. CSGC pre-diffusion

After applying the transformation Ψ , the quantum state $|C''\rangle$ can be described as follows:

$$\begin{aligned}
 |C''\rangle = \frac{1}{4}(&|00001001\rangle|0000\rangle + |11001110\rangle|0001\rangle + |11011101\rangle|0010\rangle + |01100010\rangle|0011\rangle \\
 &+ |01101111\rangle|0100\rangle + |11100001\rangle|0101\rangle + |10111011\rangle|0110\rangle + |11100111\rangle|0111\rangle \\
 &+ |11010101\rangle|1000\rangle + |00111001\rangle|1001\rangle + |01011111\rangle|1010\rangle + |00011000\rangle|1011\rangle \\
 &+ |00000110\rangle|1100\rangle + |11000010\rangle|1101\rangle + |11111010\rangle|1110\rangle + |10011101\rangle|1111\rangle)
 \end{aligned}
 \tag{8}$$

Fig. 10 depicts the quantum circuit diagram of CSGC.

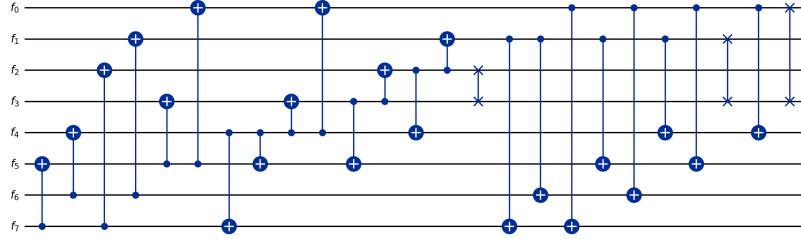


Figure 10: Quantum circuit of CSGC based pre-diffusion.

1.1.4. Quantum chaotic XOR diffusion

After applying the quantum operator U the state $|I\rangle$ can be illustrated as follows:

$$\begin{aligned}
 |I\rangle = \frac{1}{4}(&|00000000\rangle|0000\rangle + |01000100\rangle|0001\rangle + |00000111\rangle|0010\rangle + |00100001\rangle|0011\rangle \\
 &+ |00101010\rangle|0100\rangle + |10100001\rangle|0101\rangle + |01101100\rangle|0110\rangle + |00110010\rangle|0111\rangle \\
 &+ |11001100\rangle|1000\rangle + |11110000\rangle|1001\rangle + |10011111\rangle|1010\rangle + |00010110\rangle|1011\rangle \\
 &+ |10011010\rangle|1100\rangle + |10111100\rangle|1101\rangle + |01100010\rangle|1110\rangle + |11010110\rangle|1111\rangle)
 \end{aligned}
 \tag{9}$$

Fig. 11 depicts the quantum circuit diagram of XOR diffusion.

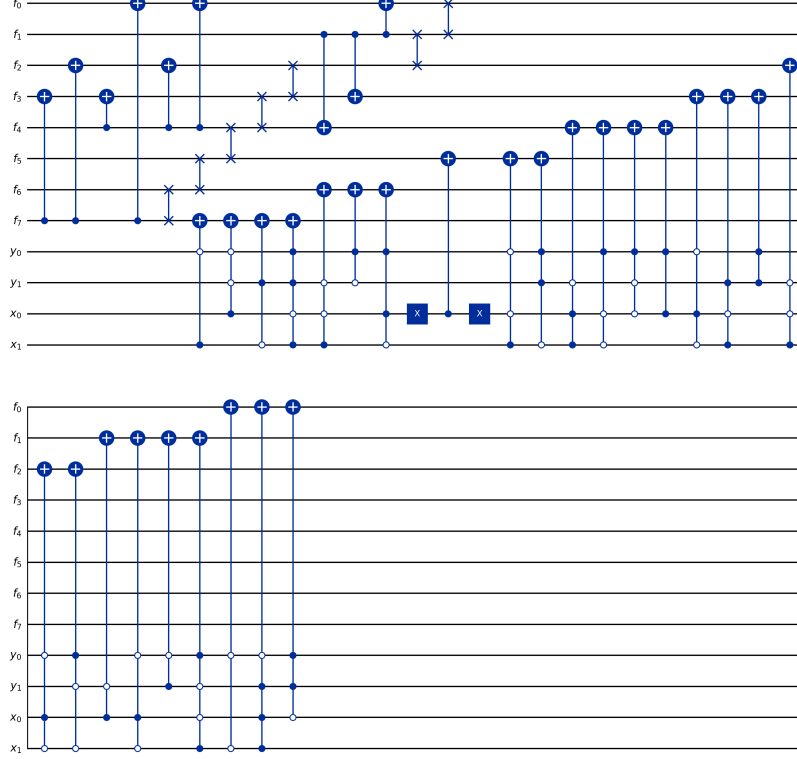


Figure 11: Chaotic XORing quantum circuit.

1.1.5. S-box substitution

The S-box operator is constructed for the S-box given in Table 6. It is then minimized using the procedure provided in Sect. 1. An ancillary register F' is introduced. Fig. 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 35, 36, 37 and 38 show the S-box circuit. The state of the system becomes $|I_{enc}\rangle$ and is described as follows:

$$\begin{aligned}
 |I_{enc}\rangle = \frac{1}{4}(&|10111001\rangle|0000\rangle + |01011111\rangle|0001\rangle + |00001010\rangle|0010\rangle + |00011100\rangle|0011\rangle \\
 &+ |01010001\rangle|0100\rangle + |01110110\rangle|0101\rangle + |00100101\rangle|0110\rangle + |01101011\rangle|0111\rangle \\
 &+ |00001110\rangle|1000\rangle + |11111101\rangle|1001\rangle + |00011101\rangle|1010\rangle + |00101110\rangle|1011\rangle \\
 &+ |01111000\rangle|1100\rangle + |00010101\rangle|1101\rangle + |00100001\rangle|1110\rangle + |00100011\rangle|1111\rangle)
 \end{aligned}
 \tag{10}$$

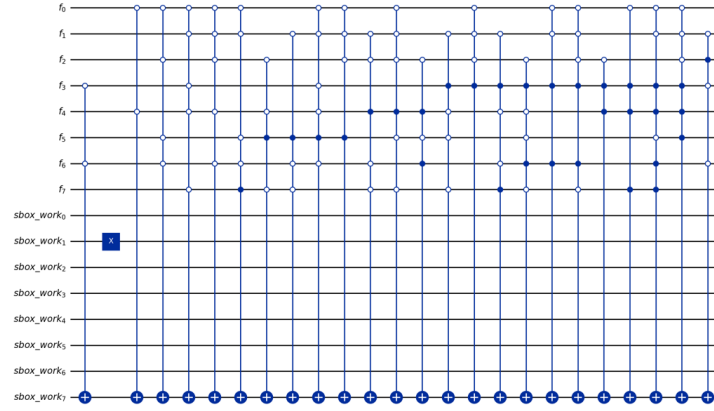


Figure 12: S-box quantum circuit

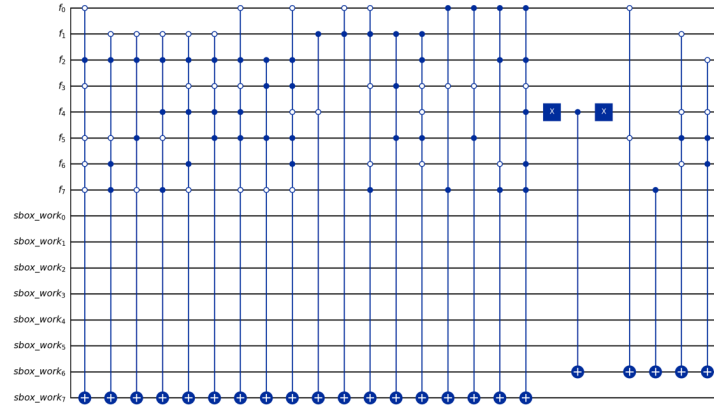


Figure 13: S-box quantum circuit

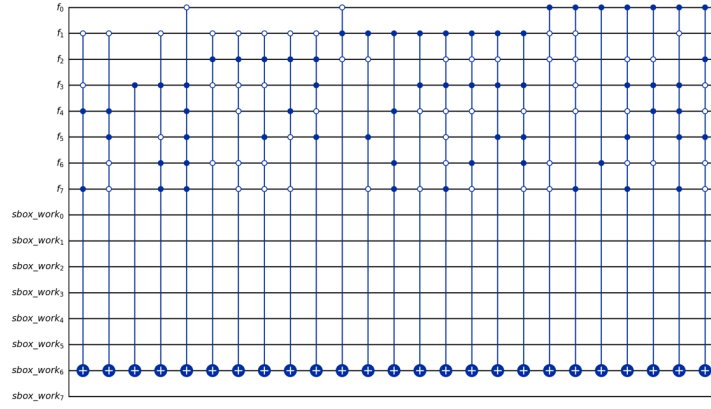


Figure 14: S-box quantum circuit

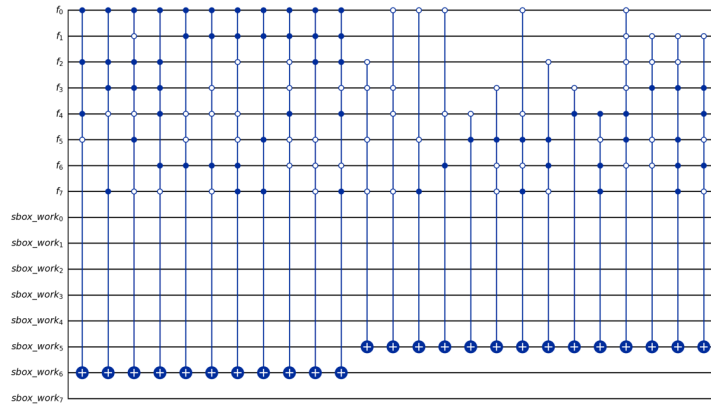


Figure 15: S-box quantum circuit (d of n)

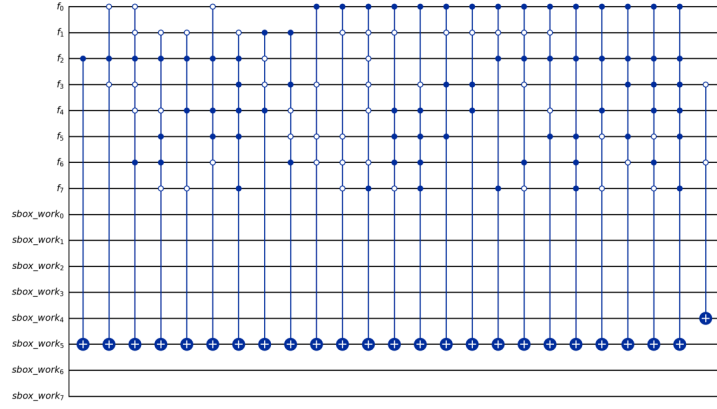


Figure 16: S-box quantum circuit

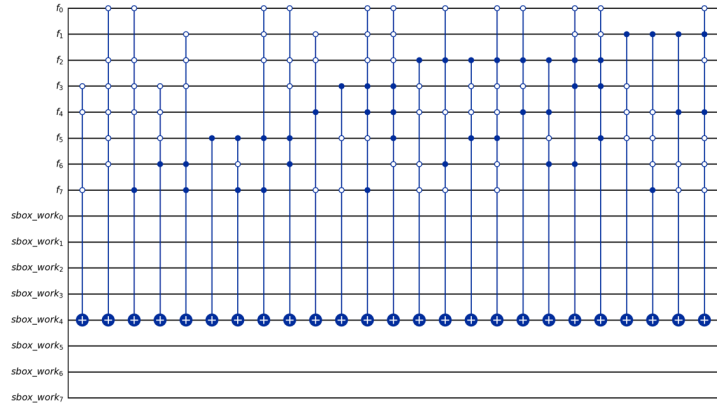


Figure 17: S-box quantum circuit

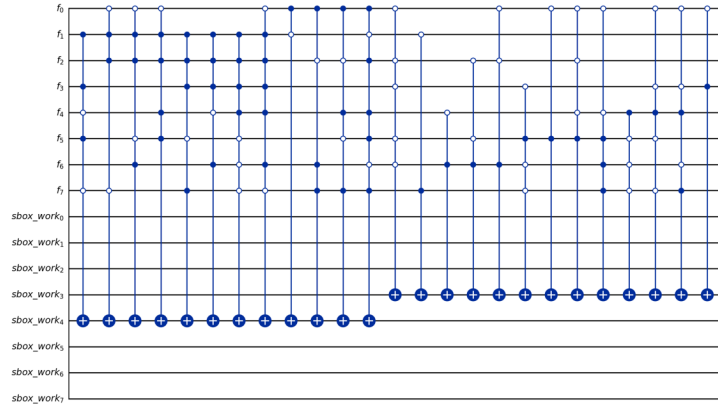


Figure 18: S-box quantum circuit

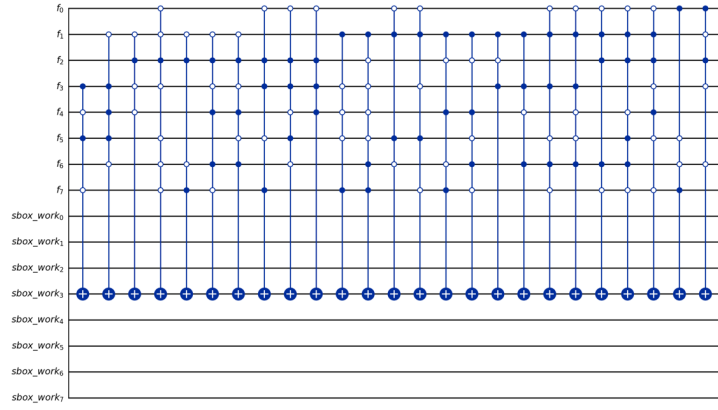


Figure 19: S-box quantum circuit

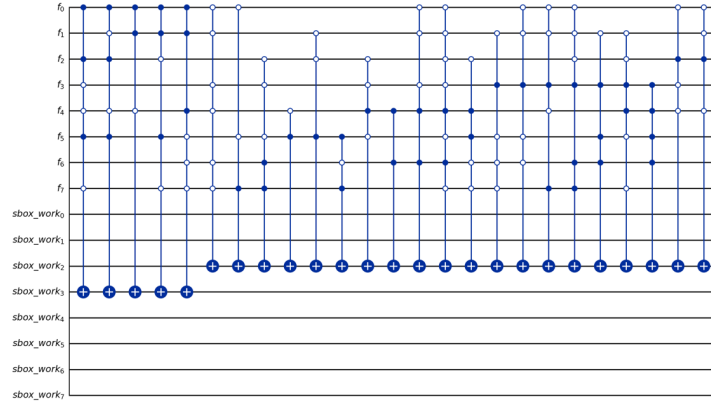


Figure 20: S-box quantum circuit

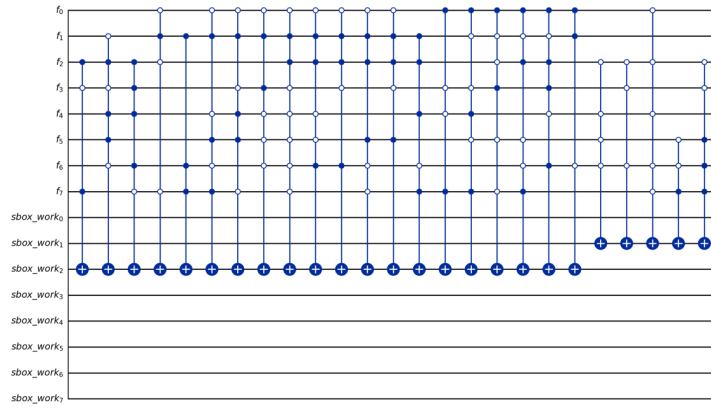


Figure 21: S-box quantum circuit

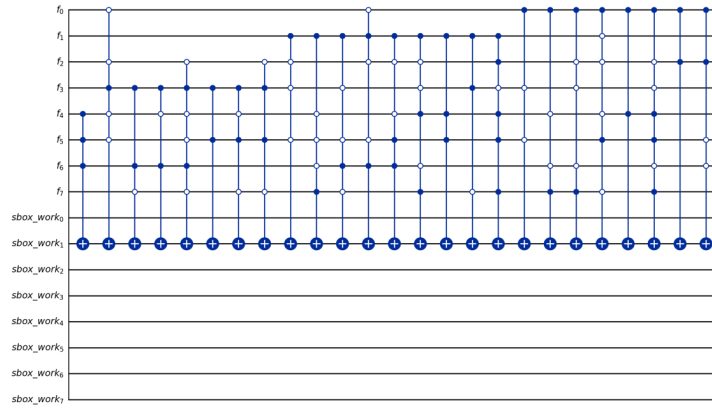


Figure 22: S-box quantum circuit

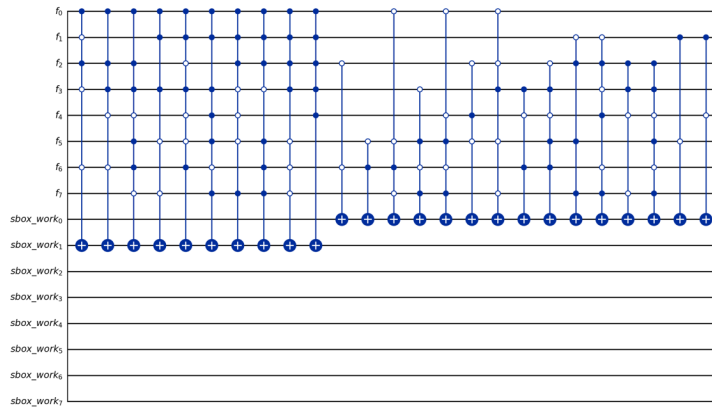


Figure 23: S-box quantum circuit

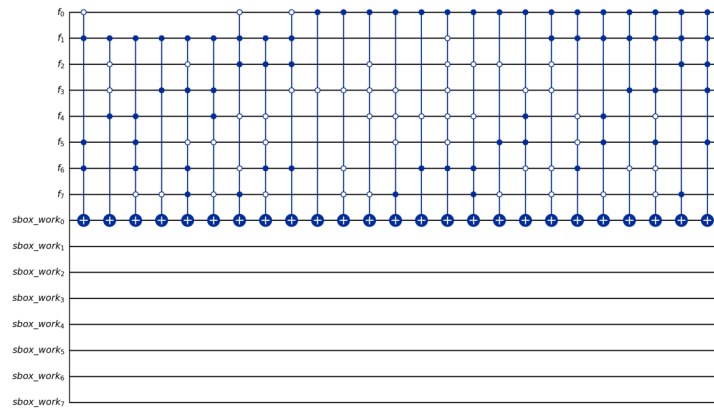


Figure 24: S-box quantum circuit

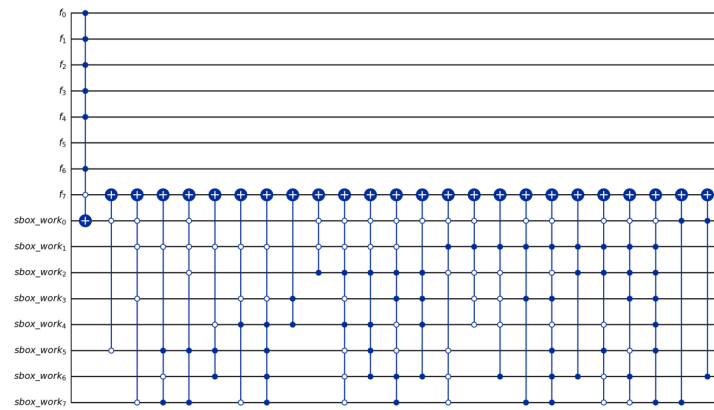


Figure 25: S-box quantum circuit (n of n)

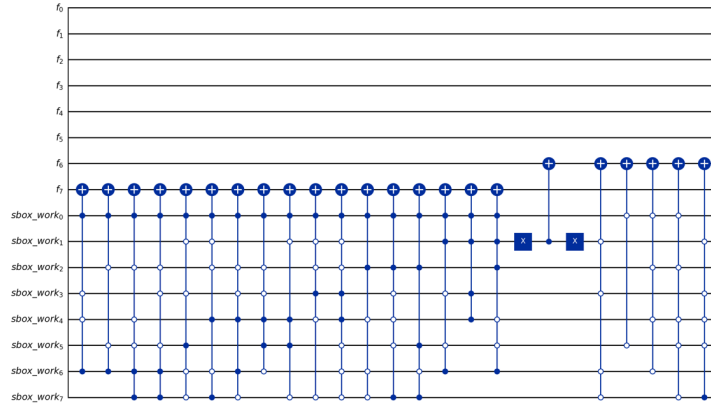


Figure 26: S-box quantum circuit

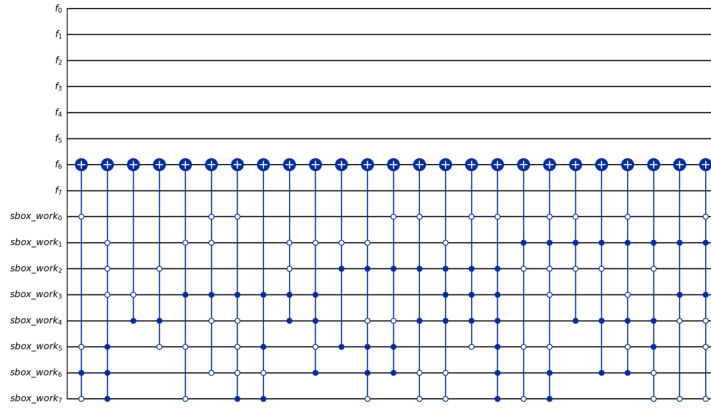


Figure 27: S-box quantum circuit

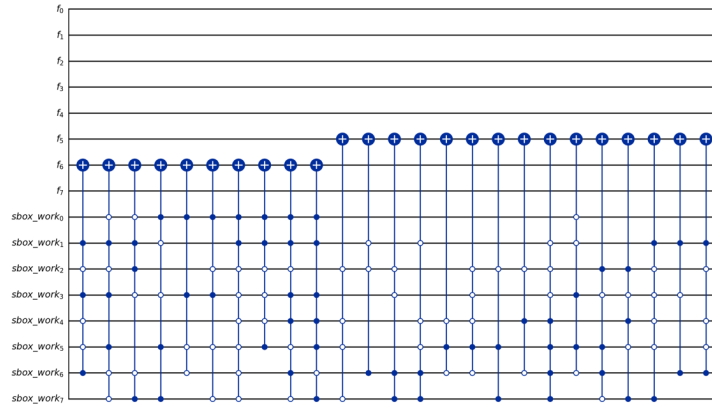


Figure 28: S-box quantum circuit

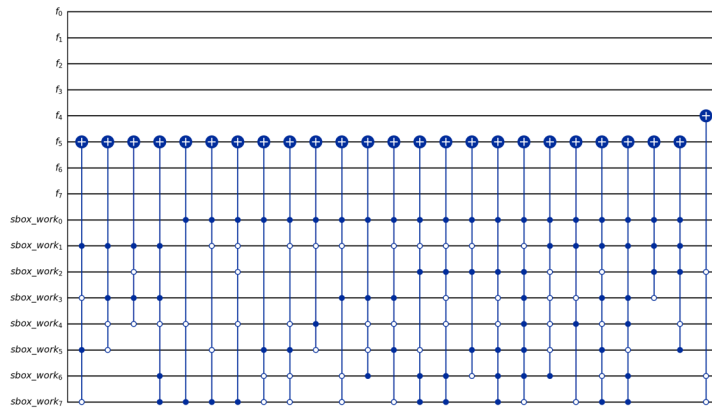


Figure 29: S-box quantum circuit

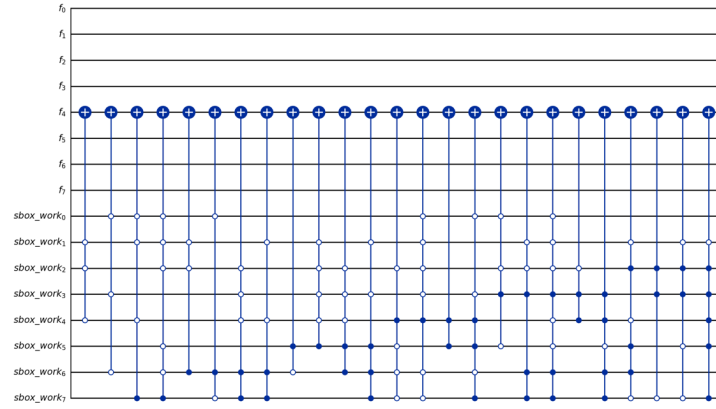


Figure 30: S-box quantum circuit

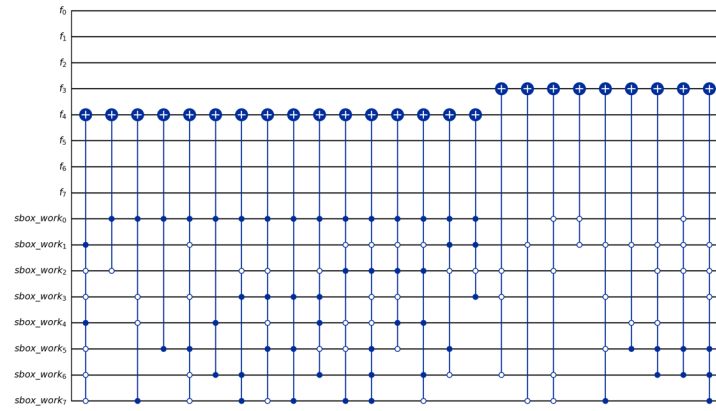


Figure 31: S-box quantum circuit

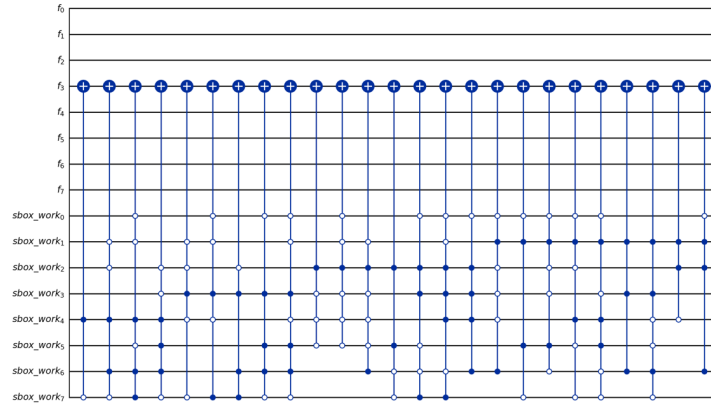


Figure 32: S-box quantum circuit

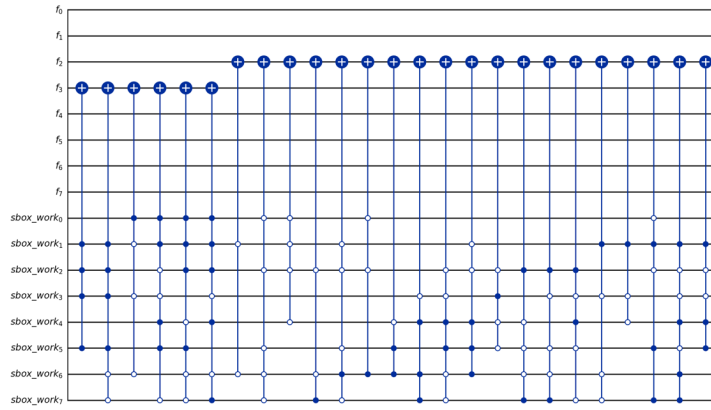


Figure 33: S-box quantum circuit

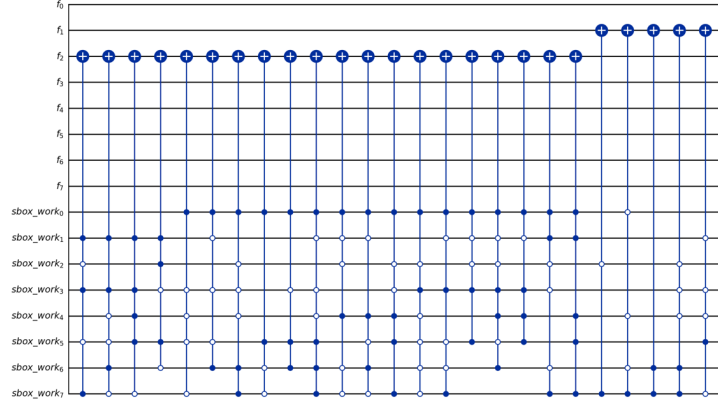


Figure 34: S-box quantum circuit

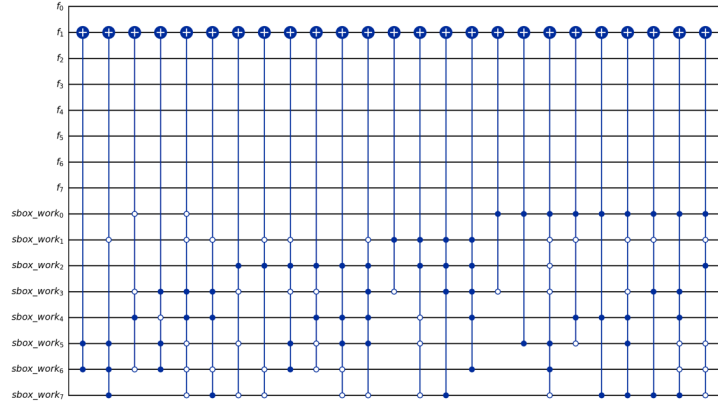


Figure 35: S-box quantum circuit

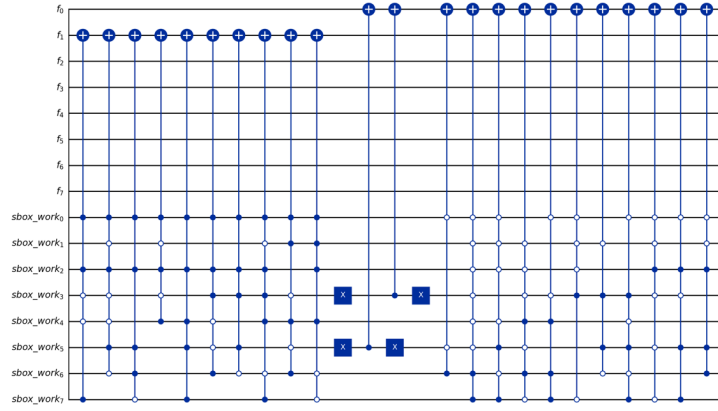


Figure 36: S-box quantum circuit

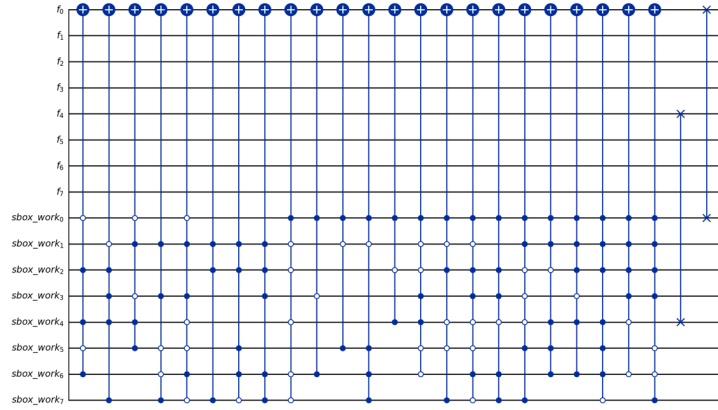


Figure 37: S-box quantum circuit

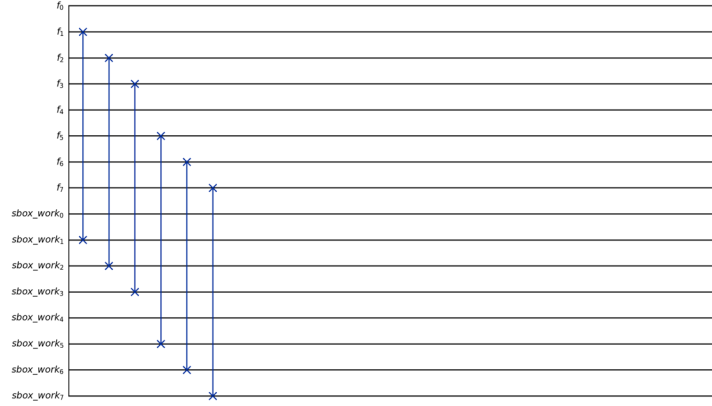


Figure 38: S-box quantum circuit

2. Security and numerical analysis

2.1. Visual and histogram analysis

This subsection computes the encryption and concealment performance of the proposed algorithm through histogram analysis, as shown in Fig. 39. The histogram of the plain image exhibits noticeable pixel distribution patterns, whereas the histogram of the encrypted image demonstrates a nearly uniform distribution [6], indicating strong diffusion characteristics and effective resistance against statistical attacks.

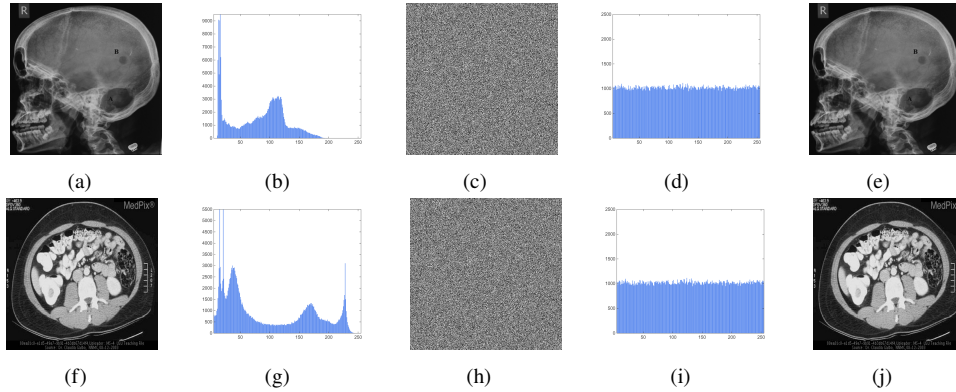


Figure 39: Histogram analysis: rows depict medical images (Brain, Cecum) and their corresponding histograms, encrypted images and their corresponding histograms, reconstructed images.

Furthermore, the (χ^2) test has been performed to validate the uniformity of the histogram distribution of the secret images [7]. A significance level of 0.05 was selected for this analysis. The mathematical formula to calculate the (χ^2) test is shown in Eq. (11):

$$\chi^2 = \sum_{i=0}^{255} \frac{(u_i - u_0)^2}{u_0}, \quad u_0 = \frac{M \times N}{256} \quad (11)$$

where u_i exhibits the frequency of the pixel value i appears in the image and u_0 depicts the expected frequency of pixel value i appears. Table 2 outcomes demonstrate that the images encrypted using the proposed algorithm successfully pass the (χ^2) -test, as all calculated (χ^2) -values are lower than the corresponding critical values.

Table 2: Evaluated (χ^2) outcome for images at $(\alpha = 0.05)$.

Theoretical Values [8]				
$\chi^2_{0.05(255)} = 293.2478$ $\chi^2_{0.01(255)} = 310.4574$ $\chi^2_{0.1(255)} = 284.3359$				
Computed metrics of the propounded algorithm				
Images	Size	Plain Images	Cipher Images	Outcome
Brain	512×512	405747.7988	196.1348	Pass
Abdomen	512×512	160421.8359	251.0098	Pass
Cecum	512×512	418530.1464	255.1094	Pass
MRI	512×512	138836.1738	243.7773	Pass
Lung1	512×512	418313.6753	208.2148	Pass
Feet	512×512	2086035.8966	240.8418	Pass
Cervical	512×512	135189.93613	232.7129	Pass

2.2. Key sensitivity analysis

Key sensitivity is an essential criterion for assessing the robustness of an encryption algorithm. In the proposed scheme, the per-image master key K_{img} is the sole secret input to the key generation procedure (Algorithm 4). To evaluate key sensitivity rigorously, each altered key is defined as an explicit single-bit flip of K_{img} ,

$$K_i = K_{img} \oplus 2^{b_i}, \quad 0 \leq b_i \leq 255, \quad (12)$$

where bit index b denotes the b -th bit of K_{img} 's 256-bit binary representation, with $b = 0$ the least significant bit. Five bit positions spanning the full key width are tested:

$$\begin{aligned}
 &\text{Correct } K_{img} = fd3e530ff58b16fac64de67c0017a7e48b2c91d3a4f76e012d9c8b5e3f1a0c7d \\
 &K_1 = K_{img} \oplus 2^0 \quad (b = 0) = fd3e530ff58b16fac64de67c0017a7e48b2c91d3a4f76e012d9c8b5e3f1a0c7c \\
 &K_2 = K_{img} \oplus 2^{64} \quad (b = 64) = fd3e530ff58b16fac64de67c0017a7e48b2c91d3a4f76e002d9c8b5e3f1a0c7d \\
 &K_3 = K_{img} \oplus 2^{128} \quad (b = 128) = fd3e530ff58b16fac64de67c0017a7e58b2c91d3a4f76e012d9c8b5e3f1a0c7d \\
 &K_4 = K_{img} \oplus 2^{192} \quad (b = 192) = fd3e530ff58b16fbac64de67c0017a7e48b2c91d3a4f76e012d9c8b5e3f1a0c7d \\
 &K_5 = K_{img} \oplus 2^{255} \quad (b = 255) = 7d3e530ff58b16fac64de67c0017a7e48b2c91d3a4f76e012d9c8b5e3f1a0c7d
 \end{aligned} \tag{13}$$

Each K_i differs from K_{img} in exactly one bit, verified by construction from Eq. (12). Fig. 40 shows the decrypted images obtained using the original key and each single-bit-perturbed key. Even a single-bit change results in complete decryption failure with no visible plain-image information. Encrypting the same image under K_{img} and each K_i independently also produces statistically uncorrelated cipher images, confirmed via absolute-difference images and histogram distributions Fig. 40. These results demonstrate the high key sensitivity and robustness of the proposed encryption scheme.

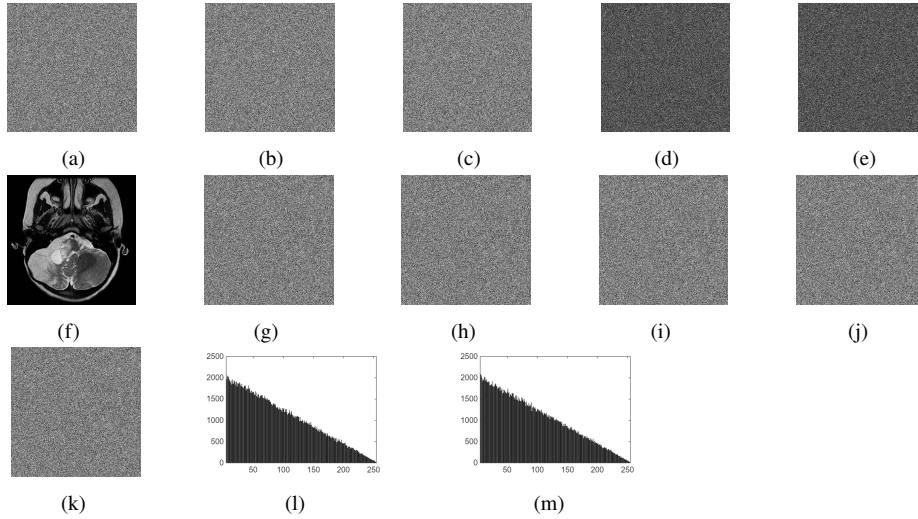


Figure 40: Key sensitivity analysis: (a)-(c) cipher images by keys K_{img} , K_1 , K_2 ; (d) $|Fig.40a - Fig.40b|$; (e) $|Fig.40a - Fig.40c|$; (f)-(k) decipher outcomes with correct and minor modified keys; (l)-(m) histogram analysis Fig.40d and 40e respectively.

2.3. Differential attack analysis

To assess the robustness of the proposed scheme against differential attacks, the *NPCR* and *UACI* are employed [8]. Their mathematical expressions are given as follows:

$$\begin{cases} \eta(r, w) = \begin{cases} 0, & \text{if } P_1(r, w) = P_2(r, w) \\ 1, & \text{if } P_1(r, w) \neq P_2(r, w) \end{cases} \\ NPCR(P_1, P_2) = \eta = \frac{\sum_{r=1}^R \sum_{w=1}^W \eta(r, w)}{R \times W} \times 100\% \end{cases} \quad (14)$$

$$UACI(P_1, P_2) = \vartheta = \frac{1}{R \times W} \left(\sum_{r=1}^R \sum_{w=1}^W \frac{|P_1(r, w) - P_2(r, w)|}{255} \right) \times 100\% \quad (15)$$

where $R \times W$ denotes the image dimensions, and $\eta(r, w)$ represents the pixel-wise difference between the two cipher images P_1 and P_2 . An image encryption scheme is considered secure if it satisfies $\eta_v > \eta_v^*$ and $\vartheta \in (\vartheta_v^{*-}, \vartheta_v^{*+})$, where v is the significance level. In this study, the proposed algorithm is evaluated at $v = 0.05$. Table 3 describes the test metrics for the images, indicating the encryption algorithm's sensitivity to minor modifications to the initial image. The computed metrics fall within the critical bounds, indicating strong resistance to differential attacks and improved performance over existing methods. Tables 4 and 5 present a comparative analysis of the *NPCR* and *UACI* values achieved by the proposed algorithm against those reported for state-of-the-art techniques.

Table 3: Comparative evaluation of $NPCR(\eta)(\%)$ and $UACI(\vartheta)(\%)$ metrics

Image	Computed metrics of η and Φ									
	Proposed		[9]		[10]		[11]		[12]	
	η	ϑ	η	ϑ	η	ϑ	η	ϑ	η	ϑ
Brain	99.591	33.463	—	—	—	—	—	—	—	—
Cecum	99.586	33.513	—	—	—	—	—	—	—	—
MRI	99.594	33.474	—	—	—	—	—	—	—	—
Barbara	99.604	33.488	99.606	33.426	99.604	33.472	99.607	33.472	—	—
Baboon	99.602	33.483	—	—	—	—	—	—	99.611	33.453
Lena	99.599	33.503	99.616	33.451	—	—	99.611	33.460	99.596	33.413
Peppers	99.603	33.495	99.607	33.432	99.606	33.459	99.608	33.466	99.612	33.477
Boat.512	99.588	33.532	—	—	99.607	33.488	99.610	33.472	99.604	33.456

Table 4: Comparative evaluation of $NPCR(\eta)(\%)$

Computed metric of η									
Image	Proposed	[9]	[10]	[11]	[12]	[13]	[14]	[15]	[16]
Brain	99.591	—	—	—	—	99.63	—	—	—
Cecum	99.586	—	—	—	—	—	—	—	—
MRI	99.594	—	—	—	—	—	—	—	—
Lung	99.607	—	—	—	—	99.67	—	—	—
Hand	99.596	—	—	—	—	99.55	—	—	—
Barbara	99.604	99.606	99.604	99.607	—	—	99.672	—	—
Baboon	99.602	—	—	—	99.611	—	99.612	—	99.28
Lena	99.599	99.616	—	99.611	99.596	—	—	—	99.39
Peppers	99.603	99.607	99.606	99.608	99.612	—	99.633	—	—
Boat.512	99.588	—	99.607	99.610	99.604	—	99.672	99.602	—
Cameraman	99.601	—	—	—	—	—	—	—	99
Woman	99.597	—	—	—	—	—	—	—	99
7.1.01	99.598	—	—	—	—	—	—	99.604	—
7.1.03	99.590	—	—	—	—	—	—	99.613	—

Table 5: Comparative evaluation of $UACI(\vartheta)(\%)$

Image	Proposed	Computed metric of ϑ							
		[9]	[10]	[11]	[12]	[13]	[14]	[15]	[16]
Brain	33.463	—	—	—	—	33.51	—	—	—
Cecum	33.513	—	—	—	—	—	—	—	—
MRI	33.474	—	—	—	—	—	—	—	—
Lung	33.439	—	—	—	—	33.53	—	—	—
Hand	33.522	—	—	—	—	33.52	—	—	—
Barbara	33.488	33.426	33.472	33.472	—	—	33.011	—	—
Baboon	33.483	—	—	—	33.453	—	33.414	—	31
Lena	33.503	33.451	—	33.460	33.413	—	—	—	32.03
Peppers	33.495	33.432	33.459	33.466	33.477	—	33.026	—	—
Boat.512	33.532	—	33.488	33.472	33.456	—	33.483	33.469	—
Cameraman	33.402	—	—	—	—	—	—	—	30
Woman	33.451	—	—	—	—	—	—	—	31.31
7.1.01	33.468	—	—	—	—	—	—	33.438	—
7.1.03	33.511	—	—	—	—	—	—	33.503	—

2.4. Parameter sensitivity and robustness analysis

The PWL-TLN experiments show that ciphertext entropy, NPCR, UACI, and pixel correlation remain relatively stable across the tested intervals. The QAT experiments cover 100 iteration counts and evaluate both post-scrambling spatial correlation and final ciphertext statistics. As expected, QAT preserves image entropy, while its spatial-correlation behavior depends on the iteration count. A one-factor-at-a-time sensitivity analysis was conducted on the 512×512 Barbara image by varying the PWL-TLN parameters b_{11} and d , GA population size N_p and generation count G_m , and QAT iteration count k . All other parameters were fixed at their baseline values $b_{11} = 0.1$, $d = 2$, $N_p = 120$, $G_m = 300$, and $k = 59$. Three repetitions were performed for each setting, and the results are presented in Figures (41)–(45). Ciphertext entropy remained close to 8 bits, while NPCR, UACI, and pixel correlation showed limited variation across the tested PWL-TLN parameter ranges. Positive finite-time Lyapunov exponents were obtained for $b_{11} \in [0.02, 0.10]$ and $d \in [1.0, 2.0]$. Similarly, variations in N_p and G_m produced only minor changes in the final ciphertext statistics. For QAT iterations $k = 1, \dots, 100$, post-scrambling spatial correlations varied with the iteration count, whereas the final ciphertext entropy, NPCR, and UACI remained relatively stable. As QAT is a permutation, the post-QAT image entropy remained unchanged. These results indicate that the encryption metrics are relatively stable within the examined parameter ranges.

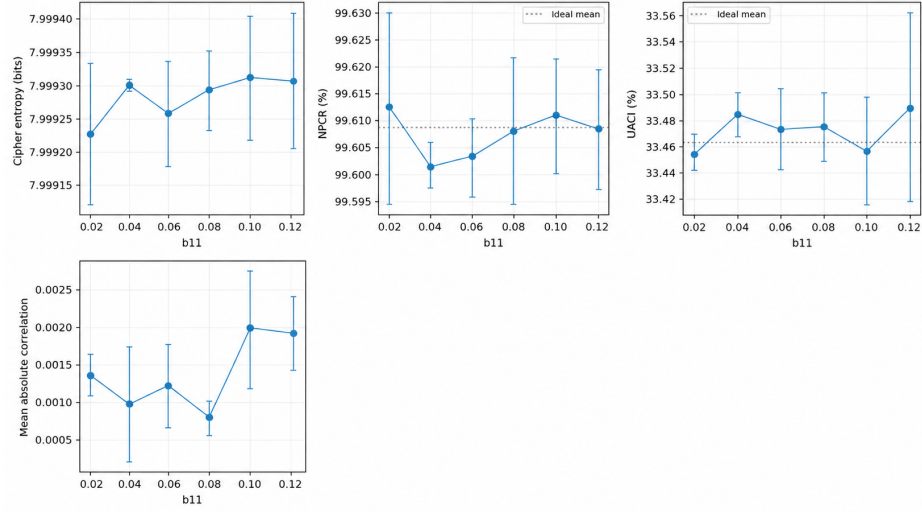


Figure 41: Variation in (a) ciphertext entropy, (b) NPCR, (c) UACI, and (d) mean absolute pixel correlation with the PWL-TLN parameter $b_1 1$.

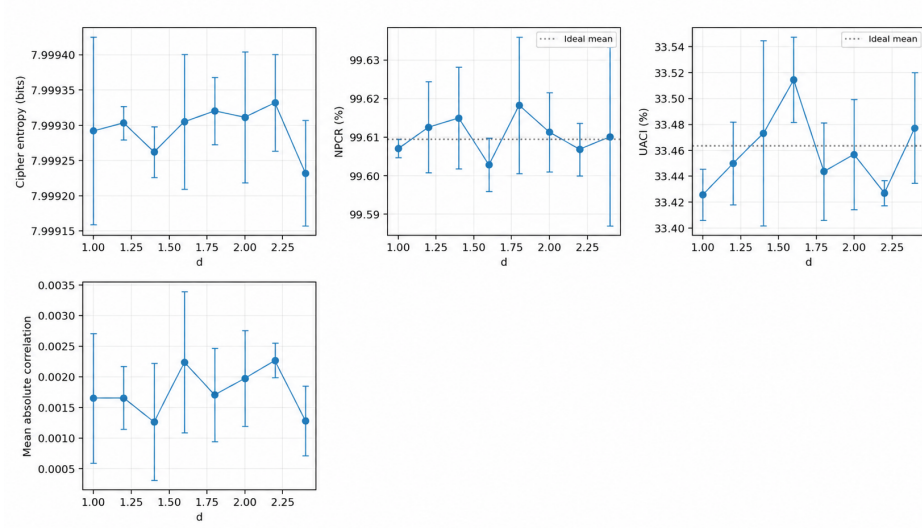


Figure 42: Variation in (a) ciphertext entropy, (b) NPCR, (c) UACI, and (d) mean absolute pixel correlation with the PWL-TLN parameter d .

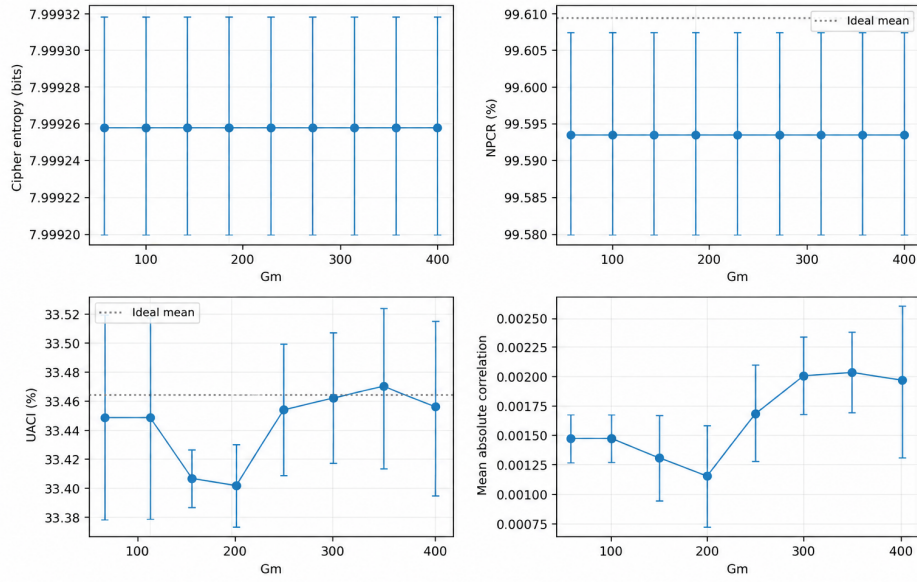


Figure 43: Variation in (a) ciphertext entropy, (b) NPCR, (c) UACI, and (d) mean absolute pixel correlation with the GA generation count G_m .

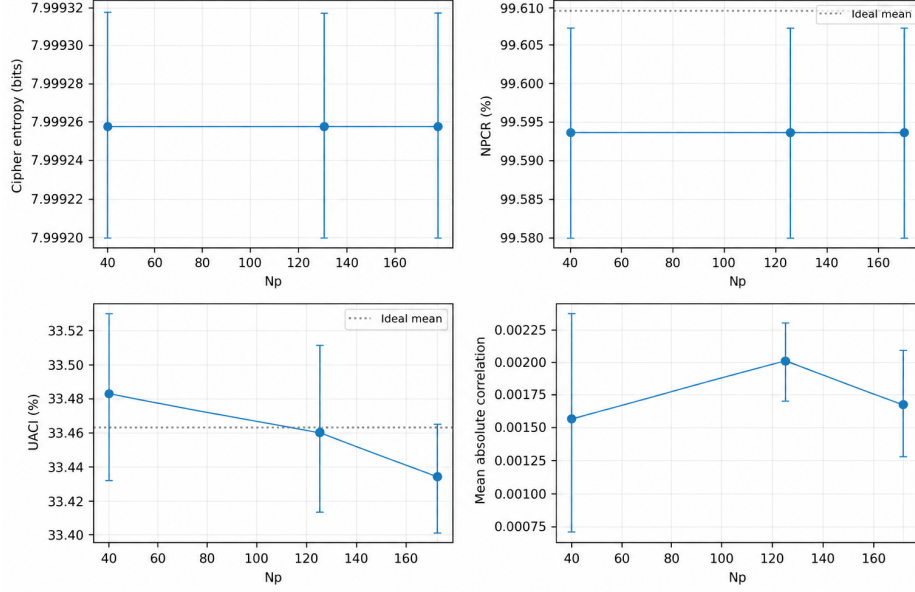


Figure 44: Variation in (a) ciphertext entropy, (b) NPCR, (c) UACI, and (d) mean absolute pixel correlation with the GA population size N_p .

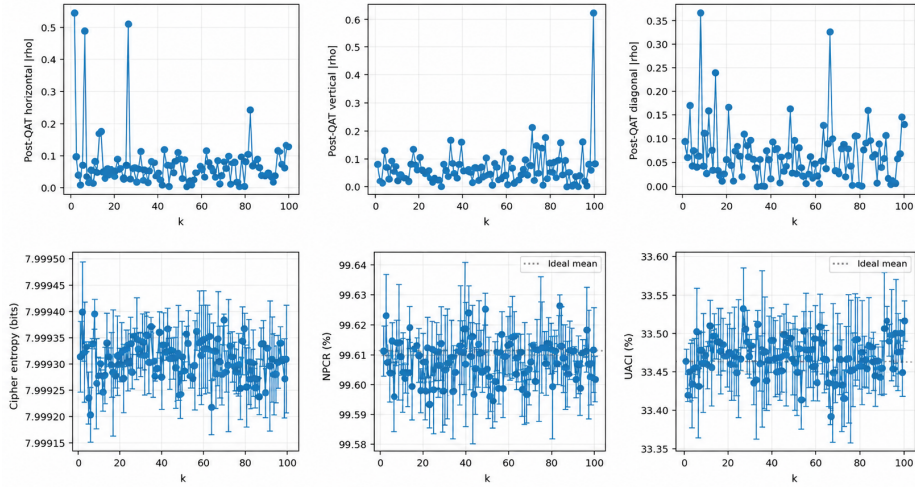


Figure 45: Variation in (a) horizontal, (b) vertical, and (c) diagonal post-QAT pixel correlations, and the final ciphertext (d) entropy, (e) NPCR, and (f) UACI with the QAT iteration count k .

2.5. Statistical analysis and security evaluation of the S-boxes

This section presents the experimental evaluation of the proposed PWLTLN-COGSA algorithm for constructing an $(8, 8)$ -bijective S-box. The generated S-box is assessed using several standard cryptographic criteria and compared with well-known S-boxes reported in the literature. The details of the evaluation metrics and the corresponding results are discussed in the following subsections. Table 6 illustrates one of the optimal S-boxes obtained using the proposed algorithm. The corresponding inverse S-box is presented in Table 7.

Table 6: An 8-bit intensity S-box.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	100	118	122	133	113	33	155	124	254	135	222	50	142	158	96	160
1	41	223	131	202	52	180	85	46	20	84	18	218	109	86	26	238
2	220	14	161	55	138	215	59	139	68	31	21	112	234	253	88	228
3	89	7	127	13	72	231	61	208	243	82	103	146	74	45	47	34
4	44	76	69	221	143	87	201	190	29	65	163	108	194	48	136	23
5	121	6	214	178	204	176	107	70	38	255	164	117	116	78	53	224
6	225	144	210	92	237	166	200	95	114	196	184	49	211	174	39	244
7	183	12	110	36	233	172	25	248	106	241	213	81	205	153	83	91
8	64	130	123	67	236	132	185	37	129	24	242	230	182	128	15	73
9	206	159	229	167	151	80	102	192	57	8	77	0	5	27	212	193
10	252	43	71	203	54	137	4	245	42	101	28	141	226	140	97	195
11	90	22	188	198	19	10	247	58	98	75	99	2	209	186	60	217
12	169	94	1	147	63	246	125	177	162	32	11	239	35	134	240	156
13	115	168	40	181	30	189	216	173	170	148	9	154	207	157	51	16
14	152	235	93	120	126	111	66	104	171	179	17	3	199	105	56	232
15	119	250	175	165	187	62	227	79	191	145	150	149	251	219	249	197

2.5.1. Nonlinearity (NL)

The nonlinearity of an S-box is mainly responsible for the non-linear relationship between the input and output vector. An S-box with low nonlinearity can be exploited using the linear cryptanalysis attack. The nonlinearity of an S-box can be measured using WHT values of all its component functions using Eq.(16). The WHT values for the COBLAH Sbox lie between $[-32, 32]$. Also, the absolute maximum Walsh coefficient value for each component function is 32, which turns the nonlinearity of each component function to 112. Tables 8 and 9 show the nonlinearity of each component function and coordinate functions of the generated S-box utilizing PWLTLN-COGSA. The Nonlinearity N_s of a S-box, S is defined as the minimum among all the hamming distances H_d between all the

Table 7: Inverse of the proposed 8-bit S-box.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	155	194	187	235	166	156	81	49	153	218	181	202	113	51	33	142
1	223	234	26	180	24	42	177	79	137	118	30	157	170	72	212	41
2	201	5	63	204	115	135	88	110	210	16	168	161	64	61	23	62
3	77	107	11	222	20	94	164	35	238	152	183	38	190	54	245	196
4	128	73	230	131	40	66	87	162	52	143	60	185	65	154	93	247
5	149	123	57	126	25	22	29	69	46	48	176	127	99	226	193	103
6	14	174	184	186	0	169	150	58	231	237	120	86	75	28	114	229
7	43	4	104	208	92	91	1	240	227	80	2	130	7	198	228	50
8	141	136	129	18	133	3	205	9	78	165	36	39	173	171	12	68
9	97	249	59	195	217	251	250	148	224	125	219	6	207	221	13	145
10	15	34	200	74	90	243	101	147	209	192	216	232	117	215	109	242
11	85	199	83	233	21	211	140	112	106	134	189	244	178	213	71	248
12	151	159	76	175	105	255	179	236	102	70	19	163	84	124	144	220
13	55	188	98	108	158	122	82	37	214	191	27	253	32	67	10	17
14	95	96	172	246	47	146	139	53	239	116	44	225	132	100	31	203
15	206	121	138	56	111	167	197	182	119	254	241	252	160	45	8	89

component functions $v.S, \in (\mathbf{F}_2^m)^*$ to the set of all affine functions ϕ .

$$N_s = \min_{\phi \in A_n} H_d(v.S, \phi) \quad (16)$$

The Nonlinearity N_s for any S-box (n, n) -function Slies between $[0, 2^{n-1} - \frac{n}{2} - 1]$. The Nonlinearity N_S of an S-box S should be as high as possible to avoid the linear approximation attack which utilizes the known plaintext to recover the cryptographic key.

Table 8: Nonlinearity of Coordinate functions of PWLTLN-COGSA S-box.

Component functions	f_1	f_2	f_3	f_4	...	...	f_{254}	f_{255}	Min	Max	Average
Nonlinearity	112	112	112	112	112	112	112	112	112	112	112

Table 9: Nonlinearity of Component functions of PWLTLN-COGSA S-box.

Component functions	f_1	f_2	f_3	f_4	f_5	f_6	f_7	f_8	Min	Max	Average
Nonlinearity	112	112	112	112	112	112	112	112	112	112	112

2.5.2. SAC analysis

SAC ensures that half of the output bit of S-Box coordinate functions gets altered when a single bit input is changed. To design a perfect S-box, the SAC should be 0.5. The SAC of an S-box can be evaluated as an average of the dependency matrix constructed utilizing the probability of avalanche value of each coordinate function computed by changing every bit of input. Table 10 presents the dependence matrix D for the PWLTLN-COGSA S-box. The SAC for the PWLTLN-COGSA S-box is found to be 0.5029, very close to the ideal value. The DSAC of the generated S-box computed using Eq.(18) is found to be 328. For all related Boolean function f , half of the bits get changed on changing a single bit input. Mathematically, the Boolean function f should satisfy:

$$f(x) \oplus f(x \oplus \gamma) = 2^{n-1} \quad (17)$$

$\forall \mathbf{F}_2^n$ and $\forall \gamma \in \mathbf{F}_2^n$ such that $H_{wt}(\gamma) = 1$. The distance to SAC(DSAC) [60] is the sum of absolute values from which a Boolean function f gets deviated from achieving the ideal value of SAC i.e. 2^{n-1} .

$$DSAC = \sum |2^{n-1} - f(x) \oplus f(x \oplus \gamma)| \quad (18)$$

For a bijective S-box S, the DSAC lies in the range $[0, 2^{n-1} * n^2]$. To satisfy the SAC, the DSAC should be as low as possible. Moreover, the offset value for the SAC can be calculated using Eq.(19):

$$SAC_{off} = \frac{1}{n^2} \left(\sum_{i=1}^n \sum_{j=1}^n |2^{-1} - D_{ij}| \right) \quad (19)$$

The offset is considered a more reliable indicator of SAC for an S-box as certain S-boxes may have an average SAC value in proximity to 0.5, and elements of the dependence matrix can significantly deviate from the ideal value of 0.5 [31]. The offset value should be as low as possible for an S-box to satisfy SAC. The COGSA S-box's offset value is 0.02002, which is very close to zero, indicating the SAC of each coordinate function is also near 0.5.

2.5.3. BIC analysis

When changing a single-bit input, BIC ensures that the output bit of S-box coordinate functions changes independently. An S-box is said to be fulfilled BIC if its coordinate functions say, $f_i, f_j; \forall i, j \in [1, n], i \neq j$ such that $f_i \oplus f_j$ satisfies

Table 10: Dependence matrix for the COGSA S-box

$i \setminus j$	f_0	f_1	f_2	f_3	f_4	f_5	f_6	f_7
f_0	0.5156	0.4531	0.5000	0.4844	0.5312	0.4688	0.5312	0.4844
f_1	0.5000	0.4688	0.4531	0.5000	0.5156	0.5312	0.4844	0.5156
f_2	0.4844	0.4844	0.4688	0.5312	0.5156	0.4844	0.5156	0.5000
f_3	0.5156	0.5000	0.4531	0.5312	0.5156	0.5156	0.5000	0.4844
f_4	0.5156	0.5469	0.5469	0.5000	0.5156	0.5000	0.4844	0.5156
f_5	0.4844	0.4531	0.4844	0.5312	0.5156	0.4844	0.5156	0.5156
f_6	0.5156	0.4688	0.5000	0.5156	0.5469	0.5156	0.5156	0.4844
f_7	0.5312	0.5469	0.5000	0.5156	0.4688	0.5156	0.4844	0.5156

the nonlinearity and avalanche criterion. To evaluate the BIC of the PWLTLN-COGSA S-box, the nonlinearity and SAC for all $f_i \oplus f_j; \forall i, j \in [1, n], i \neq j$ is computed. Tables 11 and 12 represents the BIC-SAC and BIC-NL score of the PWLTLN-COGSA S-box. The average BIC-SAC and BIC-NL score of the S-box is found to be 0.4995 and 112, respectively, which is close to the optimal S-box.

Table 11: BIC-SAC matrix for the COGSA S-box

$i \setminus j$	b_0	b_1	b_2	b_3	b_4	b_5	b_6	b_7
b_0	—	0.5137	0.5117	0.5059	0.5039	0.4766	0.4902	0.4922
b_1	0.5137	—	0.5039	0.4980	0.5078	0.4883	0.5000	0.4863
b_2	0.5117	0.5039	—	0.5000	0.5117	0.4707	0.4980	0.5000
b_3	0.5059	0.4980	0.5000	—	0.4902	0.5098	0.4922	0.5215
b_4	0.5039	0.5078	0.5117	0.4902	—	0.4941	0.5215	0.5059
b_5	0.4766	0.4883	0.4707	0.5098	0.4941	—	0.5020	0.4902
b_6	0.4902	0.5000	0.4980	0.4922	0.5215	0.5020	—	0.5000
b_7	0.4922	0.4863	0.5000	0.5215	0.5059	0.4902	0.5000	—

2.5.4. Differential cryptanalysis characteristics

Differential cryptanalysis is a chosen plaintext attack that focuses on identifying pairs of plaintexts and corresponding ciphertext sharing identical differentials. The Differential Distribution Table (DDT), constructed through Eq.(20), serves

Table 12: BIC-NL matrix for the COGSA S-box

$i \setminus j$	b_0	b_1	b_2	b_3	b_4	b_5	b_6	b_7
b_0	–	112	112	112	112	112	112	112
b_1	112	–	112	112	112	112	112	112
b_2	112	112	–	112	112	112	112	112
b_3	112	112	112	–	112	112	112	112
b_4	112	112	112	112	–	112	112	112
b_5	112	112	112	112	112	–	112	112
b_6	112	112	112	112	112	112	–	112
b_7	112	112	112	112	112	112	112	–

as a vital tool in analyzing the frequency of input and output differences $\Delta x, \Delta y$.

$$\text{DDT}(\Delta x \rightarrow \Delta y) = \#\{x \in X | S(x \oplus \Delta x) = \Delta y\} \quad (20)$$

The attacker analyzes the imbalance XOR distribution of the input and output difference present in the DDT and exploits the S-box utilizing high probability differential trials. An S-box effectively thwarts a differential cryptanalysis attack when maintaining an equiprobable input/ output XOR distribution. The complexity of the differential attack is quantified through the differential approximation probability DP, which is based on the differential approach table contains the occurrence frequency of the most probable output XOR such that:

$$DP = \max_{\Delta x \neq 0, \Delta y} \left(\frac{\#\{x \in X | S(x) \oplus S(x \oplus \Delta x) = \Delta y\}}{2^n} \right) \quad (21)$$

Table 11 presents the differential approach table for the PWLTLN-COGSA S-box. The DP for the S-box is found to be 0.01562, which is relatively low and approves the robustness of the S-box against differential cryptanalysis attacks.

2.5.5. Linear cryptanalysis characteristics

As a known plaintext attack, linear cryptanalysis exploits the highprobability occurrences of linear relations (also known as linear approximations) among plaintext, ciphertext, and cryptographic keys. To carry out this attack, the attackers utilize the Linear Approximation Table (LAT), which consists of the count of occurrences minus 2^{n-1} satisfying the linear relationship criteria for all input and output sum combinations, as expressed in Eq.(22).

$$\text{LAT}(\oplus_x \rightarrow \oplus_y) = \#\{x \in X | x \oplus_x = S(x) \oplus_y\} - 2^{n-1} \quad (22)$$

The linear probability LP, a crucial metric assessing the probability of a linear relationship between inputs and the corresponding output of an S-box, acts as a critical determinant of resilience against linear cryptanalysis. Its computation involves utilizing the Linear Approximation Table (LAT) through Eq.(23).

$$LP = \max_{\oplus_x, \oplus_y \neq 0} \left(\frac{\#\{x \in X | x \oplus_x = S(x) \oplus_y\} - 2^{n-1}}{2^n} \right) \quad (23)$$

A substitution box with a small computed LP value indicates a reduced likelihood of linear relationships between inputs and outputs, signifying a robust defense mechanism against linear cryptanalysis attacks. The LP for the S-box is 0.0625, indicating a notably low value and affirming the robustness of the proposed S-box against linear cryptanalysis attacks.

2.5.6. Comparison analysis

This subsection compares the proposed PWLTLN-COGSA S-box with several state-of-the-art S-boxes using standard cryptographic performance measures, including average nonlinearity (NL), strict avalanche criterion (SAC), SAC offset, DSAC score, BIC-SAC, BIC-NL, differential probability (DP), linear probability (LP), fixed points, opposite fixed points, and cycle structure. The comparative results are reported in Table 13. As shown in Table 13, the proposed S-box achieves an average nonlinearity of 112, which is the optimal value for an 8-bit bijective S-box and is comparable with the best reported constructions. Although some existing S-boxes also attain the same NL value, the proposed S-box shows a better overall balance across avalanche, differential, and linear properties. Its average SAC value is 0.5029, which is close to the ideal value of 0.5. Moreover, the SAC offset and DSAC values indicate that the entries of the dependence matrix remain close to their expected ideal behavior. The BIC performance further confirms the suitability of the proposed S-box for cryptographic applications. It obtains an average BIC-NL value of 112 and an average BIC-SAC value of 0.4995, showing strong independence among output bits. In addition, the proposed S-box achieves DP and LP values of 0.01562 and 0.0625, respectively, which correspond to the desirable bounds for resistance against differential and linear cryptanalysis. Furthermore, it has no fixed points or opposite fixed points and exhibits a single cycle structure. These results demonstrate that the proposed PWLTLN-COGSA S-box provides a strong and balanced cryptographic performance compared with the considered S-boxes.

Table 13: A comprehensive comparison of the proposed S-box with state-of-the-art S-boxes based on various cryptographic characteristics.

S-box	Average NL	SAC			BIC		LP	DP	FP	OFP	No. of cycles
		Average SAC	SAC offset	DSAC	Average BIC-SAC	Average BIC-NL					
[17]	108	0.5068	0.0322	528	0.5017	103.3571	0.070	0.0150	1	0	2
[18]	112	0.5017	0.0283	464	0.4591	104	0.0628	0.1171	2	0	8
[19]	112	0.5036	0.0202	332	0.50006	112	0.0625	0.01562	0	0	1
[20]	114.5	0.4975	–	–	0.50	107	0.1350	0.0391	–	–	–
[6]	112	0.5002	–	–	0.5034	112	0.0625	0.0156	0	0	1
[21]	103.5	0.5019	–	–	0.5019	103.55	0.1379	0.0443	–	–	–
[22]	112	0.4971	–	–	0.4997	112	0.0625	0.0156	0	0	1
[23]	112	0.5725	–	–	0.5011	103.36	0.1406	0.0469	–	–	–
[24]	105.4039	0.5024	–	–	0.5063	105.3571	0.1171	0.0390	–	–	–
[25]	110	0.4988	–	–	–	103.64	0.0923	0.0236	–	–	–
[26]	107.1	0.5341	–	–	0.5030	103.4	–	12	–	–	–
[27]	106	0.4983	–	–	0.5005	103.57	0.1406	0.03125	–	–	–
[28]	112	0.5734	–	–	0.5027	104	0.1328	10	2	2	–
AES	112	0.5040	0.0263	432	0.5020	112	0.0625	0.01562	0	0	1
Skipjack S-box	105.7	0.4980	0.0227	372	0.5038	112	0.0625	0.01562	0	0	1
Proposed S-box	112	0.5029	0.02002	328	0.4995	112	0.0625	0.01562	0	0	1

2.6. Cryptanalytic attack analysis

This subsection evaluates the proposed quantum image encryption framework under cryptanalytic attacks: chosen-plaintext (CPA), known-plaintext (KPA), and divide-and-conquer (DCA) attacks. The evaluation follows the actual encryption structure, consisting of GQAT scrambling, CSGC pre-diffusion, PWL-TLN-driven XOR diffusion, and block-wise S-box substitution. Let

$$C = \text{Enc}(P; K_{\text{img}}, N) \quad (24)$$

denote the encryption of plaintext image P , where

$$K_{\text{img}} \xleftarrow{\$} \{0, 1\}^{256}$$

is the per-image secret key and

$$N \xleftarrow{\$} \{0, 1\}^{96}$$

is the public nonce stored with the ciphertext metadata. The adversary $\mathcal{A}$ is assumed to know the complete encryption procedure and all public parameters but not K_{img} .

Resistance to chosen-plaintext attacks. In a chosen-plaintext attack, the adversary can submit selected images to the encryption procedure and observe their corresponding ciphertexts. Structured images such as all-black, all-white, constant-intensity, striped, checkerboard, repeated-row, repeated-column, and single-pixel impulse images are particularly useful because they may reveal fixed permutations, reused diffusion masks, or plaintext-independent substitution patterns.

For each encryption instance, the proposed key-generation procedure computes

$$\text{IKM}_i = K_{\text{img},i} \parallel h_i, \quad (25)$$

$$\text{PRK}_i = \text{HKDF-Extract}_{\text{SHA256}}(N_i, \text{IKM}_i), \quad (26)$$

followed by the domain-separated seed derivations

$$\begin{aligned} K_{\text{diff},i} &= \text{HKDF-Expand}_{\text{SHA256}}(\text{PRK}_i, \text{"PWL-TLN/diffusion-ic/v1"}, 16), \\ K_{\text{sbox},i} &= \text{HKDF-Expand}_{\text{SHA256}}(\text{PRK}_i, \text{"PWL-TLN/sbox-ic/v1"}, 16). \end{aligned} \quad (27)$$

The two seeds determine separate PWL-TLN initial states for diffusion-sequence generation and block-wise S-box selection. Under the pseudorandomness assumption of HKDF-SHA256, the use of distinct context labels computationally separates these two derived seed values.

The public nonce N_i is used only as a diversification salt. Its disclosure does not reveal the derived seeds because their confidentiality depends on the unknown $K_{\text{img},i}$. In normal operation, both $K_{\text{img},i}$ and N_i are generated independently for every encryption instance. Consequently, repeated encryption of an identical plaintext image produces different PWL-TLN trajectories, diffusion sequences, and block-wise S-box assignments.

In addition, because h_i is itself a function of P_i (and of $K_{\text{img},i}$), the derived seeds $(K_{\text{diff},i}, K_{\text{sbox},i})$ are plaintext-dependent even when K_{img} is deliberately held fixed across instances, as in the controlled diagnostic experiments below. Two distinct plaintexts $P \neq P'$ encrypted under the same (K_{img}, N) therefore yield $h \neq h'$ except with probability negligible in the collision resistance of SHA3-256, and hence $\text{IKM} \neq \text{IKM}'$ and, under the pseudorandomness of HKDF-SHA256, statistically independent derived seeds. This property is used in the known-plaintext analysis below.

Experimental verification under chosen plaintexts. The CPA evaluation comprises two complementary experiments.

First, the same plaintext image P is repeatedly encrypted using a fixed secret key K_{img} and independently generated nonces N_i :

$$C_i = \text{Enc}(P; K_{\text{img}}, N_i), \quad i = 1, 2, \dots, r, \quad (28)$$

where $N_i \neq N_j$ for $i \neq j$. This is a conservative test because the normal operating procedure renews both K_{img} and N . The test isolates the diversification introduced by the nonce and verifies that repeated encryption of the same image does not reuse an internal diffusion or substitution pattern. Successful non-repetition requires

$$C_i \neq C_j, \quad \forall i \neq j. \quad (29)$$

Pairwise NPCR, UACI, and bit-level Hamming distance are used to quantify the separation among the resulting ciphertexts. Cryptographic hash values may additionally be reported as a basic check that no two ciphertexts are identical. Second, the adversary-selected plaintext set is defined as

$$\mathcal{P}_{\text{CPA}} = \{P_{\text{black}}, P_{\text{white}}, P_{\text{gray}}, P_{\text{row}}, P_{\text{column}}, P_{\text{stripe}}, P_{\text{checker}}, P_{\text{impulse}}\}. \quad (30)$$

Each structured plaintext is encrypted using the proposed per-encryption key-generation procedure. The resulting cipher images are evaluated through histogram uniformity, information entropy, adjacent-pixel correlation, and visual inspection. A secure response should reveal no visible structure corresponding to the selected plaintext patterns and should not expose a reusable permutation, diffusion mask, or block-wise substitution pattern. To evaluate the statistical resilience of the scheme under a CPA scenario, five different images of size 512×512 were encrypted. The resulting cipher images exhibit near-uniform histograms and negligible pixel correlation, as shown in Fig. 46 and Table 14. These results confirm the strong randomness of the ciphertext and the scheme's effective resistance to both known-plaintext and chosen-plaintext attacks. The GQAT stage redistributes pixel positions, CSGC pre-diffusion increases intra-pixel dependency, PWL-TLN-driven XOR diffusion propagates local changes across the image, and the block-wise S-box stage introduces non-linear substitution. Therefore, the selected structured plaintexts pass through both key-dependent diffusion and key-dependent substitution before producing the final ciphertext.

Differential chosen-plaintext analysis. Differential analysis evaluates the sensitivity of the encryption transformation to a minimal plaintext modification. Let

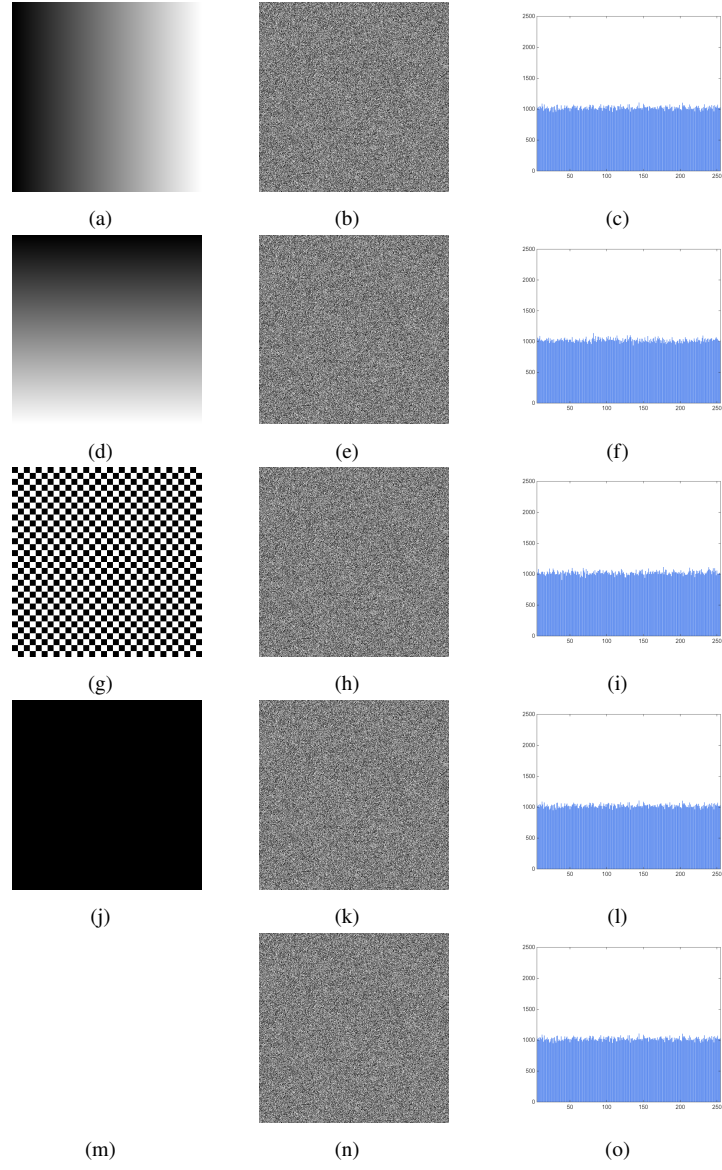


Figure 46: Histogram analysis: first column depict different images, second column present corresponding encrypted images and third column illustrate the corresponding histograms.

Table 14: Statistical metrics of encrypted uniform images.

Image	Entropy	Correlation coefficient		
		H	V	D
P_{black}	7.9993	-0.0051	-0.0111	0.0041
P_{white}	7.9992	-0.0013	-0.0019	-0.0036
P_{gray}	7.9992	-0.0025	-0.0045	-0.0051
$P_{checker}$	7.9993	-0.0149	0.0003	0.0412
P_{row}	7.9994	-0.0064	-0.0021	0.0006
P_{column}	7.9993	0.0003	-0.0189	0.0005

P_1 and P_2 differ in only one pixel or one bit. To isolate plaintext sensitivity, both images are intentionally encrypted using the same K_{img} and N :

$$C_1 = \text{Enc}(P_1; K_{img}, N), \quad (31)$$

$$C_2 = \text{Enc}(P_2; K_{img}, N). \quad (32)$$

Although the operational scheme renews K_{img} and N for each encryption instance, both values are deliberately fixed in this controlled diagnostic experiment, isolating the propagation induced by the plaintext modification itself. Because h is incorporated into IKM (Algorithm 4), fixing K_{img} and N does not fix the derived seeds: $P_1 \neq P_2$ generally yields $h_1 \neq h_2$ and hence distinct (K_{diff}, K_{sbox}) pairs, so C_1 and C_2 generally differ in diffusion sequence and block-wise S-box assignment, in addition to the deterministic propagation through GQAT and CSGC. The measured NPCR and UACI values therefore reflect both this key-schedule-level divergence and the structural avalanche of the diffusion and substitution stages. For two cipher images of size $H \times W$, the number of pixels change rate is computed as

$$\text{NPCR} = \frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W D(i, j) \times 100\%, \quad (33)$$

where

$$D(i, j) = \begin{cases} 0, & C_1(i, j) = C_2(i, j), \\ 1, & C_1(i, j) \neq C_2(i, j). \end{cases} \quad (34)$$

The unified average changing intensity is defined as

$$\text{UACI} = \frac{1}{255HW} \sum_{i=1}^H \sum_{j=1}^W |C_1(i, j) - C_2(i, j)| \times 100\%. \quad (35)$$

For independent uniformly distributed 8-bit cipher pixels, the theoretical reference values are approximately

$$\text{NPCR}_{\text{ideal}} = 99.6014\%, \quad (36)$$

$$\text{UACI}_{\text{ideal}} = 33.4635\%. \quad (37)$$

Values close to these references indicate that a one-pixel modification in the plaintext propagates over nearly the entire cipher image. This experiment measures the intrinsic diffusion and avalanche behavior of the encryption transformation and does not represent the normal per-encryption key-renewal policy.

Resistance to known-plaintext attacks. In a known-plaintext attack, the adversary is assumed to possess one or more plaintext–ciphertext pairs (P, C) together with the public parameters and metadata, including the nonce N , plaintext-binding digest h , GQAT and CSGC parameters, the PWL-TLN model, and the public S-box pool. The objective is to recover the master key K_{img} , the derived control sequences, or an equivalent decryption component. For a known plaintext P , the output of the public GQAT and CSGC stages can be computed as

$$I_2 = \text{CSGC}(\text{GQAT}(P)). \quad (38)$$

The remaining encryption relation is

$$C = \text{Sub}_{\sigma}(I_2 \oplus \mathcal{D}), \quad (39)$$

where $\mathcal{D}$ is the PWL-TLN-derived diffusion sequence and σ denotes the block-wise S-box assignments. Both quantities are derived from the secret master key through the domain-separated HKDF procedure.

For any guessed assignment $\hat{\sigma}$, an algebraically consistent candidate diffusion sequence can be obtained as

$$\hat{\mathcal{D}} = I_2 \oplus \text{Sub}_{\hat{\sigma}}^{-1}(C). \quad (40)$$

Therefore, a single known plaintext–ciphertext pair does not by itself identify the genuine pair $(\mathcal{D}, \sigma)$, since different candidate S-box assignments can be accompanied by corresponding diffusion sequences satisfying Eq. (39). Distinguishing the genuine pair would additionally require verifying that the candidate

sequences are consistent with the secret-key derivation and PWL-TLN generation process. The derived diffusion and substitution controls are instance dependent. In the proposed key-generation procedure, they are obtained from K_{img} , the nonce N , and the plaintext-dependent digest h . Consequently, plaintext-ciphertext pairs generated under different encryption instances do not normally reuse the same diffusion sequence or block-wise S-box assignment. Thus, recovery of an equivalent $(\mathcal{D}, \sigma)$ pair for one instance would not directly provide a reusable decryption component for another instance. The public plaintext-binding digest may also be used to verify a candidate master key in a known-plaintext setting, but under the assumed preimage resistance of SHA3-256 it does not provide a practical shortcut for recovering the 256-bit secret key. Under the pseudorandomness assumptions of HKDF-SHA256, no efficient method for deriving the secret control sequences from the public information alone is identified. Hence, the generic root-key search remains the baseline key-recovery attack, with complexity $O(2^{256})$ classically and order $O(2^{128})$ under an ideal Grover search, as discussed in Section 4.5.

Resistance to divide-and-conquer attacks. A divide-and-conquer attack attempts to reduce the cryptanalytic effort by separating the encryption system into smaller independently solvable components. In the proposed scheme, an adversary may attempt to recover the diffusion sequence $\mathcal{D}$ and the block-wise S-box assignments σ separately.

The derivation chain is

$$\begin{aligned}
\text{IKM} &= K_{\text{img}} \parallel h, \\
\text{PRK} &= \text{HKDF-Extract}_{\text{SHA256}}(N, \text{IKM}), \\
K_{\text{diff}} &= \text{HKDF-Expand}_{\text{SHA256}}(\text{PRK}, \text{"PWL-TLN/diffusion-ic/v1"}, 16), \\
K_{\text{sbox}} &= \text{HKDF-Expand}_{\text{SHA256}}(\text{PRK}, \text{"PWL-TLN/sbox-ic/v1"}, 16), \\
K_{\text{diff}} &\longrightarrow \mathbf{x}^{(d)}(0) \longrightarrow \mathcal{D}, \\
K_{\text{sbox}} &\longrightarrow \mathbf{x}^{(s)}(0) \longrightarrow \sigma.
\end{aligned} \tag{41}$$

The distinct HKDF context labels prevent direct derivation of K_{sbox} from K_{diff} , or vice versa, under the pseudorandomness assumption of HKDF-SHA256. Nevertheless, domain separation alone does not prove that both branches must always be recovered through one joint search.

Table 15 summarizes the public and key-dependent components of the proposed scheme.

Table 15: Public and key-dependent components of the proposed encryption scheme.

Component	Source	Key-dependent
Nonce N	Metadata Φ	No
Digest h	Metadata Φ	No
Diffusion $\mathcal{D}$	K_{diff}, N	Yes
S-box indices $\{\sigma_b\}$	K_{sbox}, N	Yes

Attempted recovery of the S-box assignments. For an image divided into B blocks, each block is assigned an S-box from the public pool $\mathcal{P}$. Although the apparent assignment space

$$|\mathcal{P}|^B$$

may be very large, it must not be interpreted as independent secret-key entropy because all assignments are deterministically generated from K_{sbox} .

A candidate assignment cannot be distinguished from an incorrect one using the ciphertext alone: as established in the known-plaintext analysis above, every candidate $\hat{\sigma}$ admits an algebraically consistent $\hat{\mathcal{D}}$, so satisfying the observed relation is not evidence of correctness. Therefore, the S-box assignment-space size is not used as an additional key-space claim.

Attempted recovery of the diffusion sequence. For the XOR diffusion rule

$$I_3 = I_2 \oplus \mathcal{D}, \quad (42)$$

a guessed block-wise assignment $\hat{\sigma}$ allows the adversary to compute

$$\hat{I}_3 = \text{Sub}_{\hat{\sigma}}^{-1}(C) \quad (43)$$

and an algebraically consistent candidate diffusion mask

$$\hat{\mathcal{D}} = I_2 \oplus \hat{I}_3. \quad (44)$$

This observation shows the public S-box pool are not treated as independent key entropy: for any guessed $\hat{\sigma}$ there exist a $\hat{\mathcal{D}}$. Even if an equivalent diffusion mask were recovered under a strong known-plaintext assumption, it would apply only to that encryption instance. Since K_{img} , N , and h are freshly determined for every encryption, the recovered mask does not reveal the root key or provide a reusable mask for any other ciphertext.

Coupling between diffusion and substitution. The diffusion sequence $\mathcal{D}$ and S-box assignment σ are domain-separated in derivation (distinct HKDF context labels) but originate from the same root secret K_{img} via a common PRK. As shown above, the two unknowns cannot be separated using the ciphertext relation

$$I_{\text{enc}} = S_{\sigma_b} \left(\text{Diff}_{\mathcal{D}}(I_2) \right) \quad (45)$$

alone: this single equation is satisfied by every candidate σ together with its algebraically determined $\hat{\mathcal{D}}$. No efficient distinguisher for rejecting incorrect candidates is established in this work. In its absence, recovering the genuine $(\mathcal{D}, \sigma)$ pair reduces to recovering the root key K_{img} from which both are deterministically derived via the HKDF chain of Algorithm 4. The corresponding brute-force complexity is therefore

$$|\mathcal{K}| = 2^{256} \quad (46)$$

classically, reducing to approximately 2^{128} under Grover’s algorithm (Section 4.5) the cost of exhaustively searching K_{img} directly, not a joint search over $(\mathcal{D}, \sigma)$ treated as independent objects. The results of this analysis indicate that the proposed scheme resists divide-and-conquer decomposition of $(\mathcal{D}, \sigma)$ under the stated assumptions: pseudorandomness of HKDF-SHA256, preimage and collision resistance of SHA3-256, and the absence of a proven efficient distinguisher for candidate S-box assignments.

Remark (seed non-reuse across instances). Because K_{img} and N are freshly sampled for every encryption and h additionally binds the derived seeds to P , no two encryption instances observed by the adversary share $(\mathcal{D}, \sigma)$ except with probability negligible in the security parameters of SHA3-256 and the CSPRNG used to sample K_{img} and N . In particular, the scheme is not vulnerable to cross-ciphertext attacks that exploit reused diffusion masks or substitution patterns (e.g., XOR-cancellation attacks across two ciphertexts sharing $\mathcal{D}$), since such reuse would itself require a (K_{img}, N, h) collision.

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